



Solid coffee waste as alternative to produce carotenoids with antioxidant and antimicrobial activities

Mariana Dias Moreira^{a,b}, Marcela Magalhães Melo^a, Jéssica Marques Coimbra^a, Kelly Cristina dos Reis^b, Rosane Freitas Schwan^a, Cristina Ferreira Silva^{a,*}

^a Department of Biology, Federal University of Lavras, Lavras, MG, Brazil

^b Department of Chemistry, Biotechnology and Bioprocess Engineering, Federal University of São João Del Rei, Campus Alto Paraopeba (CAP), OuroBranco, MG, Brazil

ARTICLE INFO

Article history:

Received 25 May 2018

Revised 8 October 2018

Accepted 10 October 2018

Keywords:

Pigments

Rhodotorula

Waste coffee

Biological activity

ABSTRACT

Special coffee production involves particular sensorial characteristics of the beverage as well as the production system of coffee beans, in particular environmental issues like water and solid waste disposal. Many countries around the world have problems with that waste disposal. The possibility to integrate the commodity production with suitable agricultural practices was the focus on this work using the solid waste to produce pigments by yeasts with biological activities. The better carotenoids production was tested previously in husk and pulp extract. The production of total carotenoids by yeast was $16.36 \pm 0.073 \text{ mg L}^{-1}$ was $21.35 \pm 0.067 \text{ mg L}^{-1}$, in the pulp extract and husk extract respectively. Carotenoids produced, exhibited antioxidant and antimicrobial activities against pathogenic bacteria such as *Salmonella colorless*, *Escherichia coli*, *Staphylococcus aureus* and *Listeria monocytogenes* as well as toxigenic fungi like *Aspergillus flavus*, *A. parasiticus*, *A. carbonarius* and *A. ochraceus*. These characteristics of the pigments are important to replace the artificial ones commonly used in food and pharmaceutical industries allowing the consumers to choose more natural products at lower costs.

© 2018 Published by Elsevier Ltd.

1. Introduction

The increasing search by consumers for natural products has influenced the world market to develop more natural and healthy products with nutraceutical properties (Nasri et al., 2014), thus guaranteeing the maintenance of health. However, the first criterion used in choosing a product is its appearance, especially color. Generally, there is addition of artificial dyes which are admittedly harmful to health because they can be accumulative and toxic to the human body and damage our surrounding environment (Malik et al., 2012). In this context, the natural food colors market is targeting the needs of consumers by increasingly replacing synthetic additives with natural additives (Kaiser et al., 2007; Malik et al., 2012).

Therefore, the search for natural dyes, mainly of microbial origin, has attracted the interest of many researchers since they can be produced in a short period of time and at any time of year (Valduga et al., 2008). Besides being a natural product, the biological relevance of the carotenoid is related to potential antioxidant

and antimicrobial activities and that is the reason for their wide application in the food and pharmaceutical industries (Kirti et al., 2014). Interest in carotenoids has increased in recent years due to its proven pro-vitamin A activity and evidence of other biological properties such as antioxidant and anticarcinogenic properties, prevention of macular degeneration and cardiovascular diseases (Maldonado et al., 2008; Mata-Gómez et al., 2014).

β -carotene and astaxanthin are the natural pigments that are mostly used in industries as dyes for food and animal feed and they are produced by strains of the fungus *Blakesleatrispora*, microalgae as *Dunaliella*, *Haematococcus* sp. and *Phaffiarhodozyma* yeast (Malik et al., 2012). However, the production of microbial pigments is still limited by the high cost of the production process. An alternative to lower the cost of production is the use of wastes generated during the food production process (Aksu and Eren, 2005; Marova et al., 2012). Therefore, combining the production of foods with more natural ingredients with the use of residues generated in the food production turns agro-industrial waste into an attractive substrate.

The types and amounts of carotenoids produced may vary depending on the carbon and nitrogen sources, minerals, vitamins, pH, aeration, temperature, light and stress (Aksu and Eren, 2005; Valduga et al., 2009). The production of carotenoids from different

* Corresponding author at: Department of Biology, Federal University of Lavras, CEP: 37.200-000, Lavras, MG, Brazil.

E-mail address: cristinafsb@dbi.ufla.br (C.F. Silva).

substrates has been reported in some papers through the use of grains (oats, wheat, barley, maize, rice, rye), lipids, glycerol, cellobiose, shrimp waste, sugarcane, molasses, grape juice, cheese and whey milk from different microbial strains (Marova et al., 2012; Sachindra et al., 2006; Valduga et al., 2009).

The production of carotenoids by microorganisms using agricultural waste has been reported for at least 15 years, but this is the first report regarding the use of coffee pulp and husk to produce carotenoids using non-conventional yeast. This investigation shows the possibility to integrate the commodity production with environment-friendly practices.

2. Experimental

2.1. Raw materials and pretreatment

Coffee husk and pulp from dry and semi-dry process, respectively, of the coffee beans (*Coffea arabica* L., variety red Catuaí 99), were supplied by a coffee-producing unit located in the southern area of the state of Minas Gerais (Brazil). The solid waste was immediately frozen at -20°C . Before the use, the husk and pulp coffee were dried at 65°C , separately, until they reached a constant weight (AOAC) (1995), and ground in Willey mill (1.0 mm) and stored in plastic flasks. Before the use as substrate, the residues were pre-treated. The pretreatment was done to increasing of the pH value of wastes. During the pretreatment process, each substrate (50 g) was resuspended in 1 L of distilled water containing KOH 0.06% (w/v) and sterilization was carried out at $121^{\circ}\text{C}/30\text{ min}$ (Machado et al., 2002). The extracts were filtered with Whatman N° 1 and used for fermentation media. The solid fraction was discharged.

2.2. Physicochemical analysis

Husk and pulp coffee (*in natura* and pretreated extract) were characterized according to their crude protein content, their total soluble carbohydrates and pH. The determination of crude protein and the contents of the dry matter (DM) definitive were carried out according to the methods recommended by the Association of Official Agricultural Chemists (AOAC, 1995). The total soluble carbohydrates were determined by phenol-sulfuric method described by Dubois et al. (1956). The pH was determined by using a digital potentiometer (Digimed analytical, Model DM22).

2.3. Microorganism and fermentation conditions

The strain used in this study, *Rhodotorula mucilaginosa* CCMA 0156, belongs to the Culture Collection of Agricultural Microbiology (CCMA) of the Biology Department, Federal University of Lavras, Brazil. This strain was previously selected to produce higher biomass production in pulp and husk coffee as substrate.

The initial culture was reactivated in 1 mL of Yeast Malt Extract media (YM) (g L^{-1}) (w/v): 3.0 yeast extract, 3.0 malt extract, 5.0 peptone and 10.0 glucose, $28^{\circ}\text{C}/48\text{ h}$. After the time incubation, the contents were added to a tube filled with 9 mL YM for 24 h. The same was carried out in Erlenmeyer flasks of 250 mL with 90 mL of the same media but which were incubated for 48 h. Before inoculation of the fermentation media with the solid waste, the cells were centrifuged at $9000 \times g$, $4^{\circ}\text{C}/10\text{ min}$ and washed 2 times with distilled water. The starter culture was 10.0% (v/v) and the process was done by batch fermentation. The production of carotenoids was performed in two different media. The first contained coffee pulp extract (6.68% w/v), peptone (10.04 g L^{-1}), yeast extract (3 g L^{-1}), glucose (2 g L^{-1}) and Tween 80 (0.5% w/v). The second media contained coffee husk extract (8.36% w/v), glucose

(6.36 g L^{-1}), peptone (3.68 g L^{-1}) and Tween 80 (0.5% w/v). The difference in the composition of the two media and the conditions used were determined in previous studies in optimization condition (data not shown) were just peptone and the husk and pulp extract showed statistical difference. The production of carotenoids was carried out in 500 mL Erlenmeyer flasks containing 300 mL of media, inoculated with 10^7 cell mL^{-1} and incubated at 28°C , 160 rpm for 5 days in the dark. The experiments were conducted in triplicate.

2.4. Recovery and quantification of carotenoids

The recovery of total carotenoids was carried out according to the methodology described by Valduga et al. (2009), with modification using acetone: methanol solvent solution (7:3, v/v). Previously, biomass recovered from the fermentation medium was centrifuged ($9,000 \times g$, 4°C , 10 min) and dried in oven at $35^{\circ}\text{C}/24\text{ h}$. After that, the biomass dried was macerated in liquid N_2 and added to 2 mL of dimethyl sulfoxide (DMSO) and heated in water bath at $55^{\circ}\text{C}/30\text{ min}$. After this time, 2 mL of mixed solvent were added and centrifuged ($5,000 \times g$, 4°C , 10 min). The supernatant was then separated, and successive extractions were carried out until both solvent and cells remained colorless. The solvent solution was evaporated over atmospheric nitrogen gas and the pigments were solubilized in 5 mL methanol (99%) solution.

According to the equation described by Davies (1976), the concentration of total carotenoids was estimated by the maximum absorbance at 450 nm. The concentration was expressed in total carotenoids ($\mu\text{g L}^{-1}$) and specific carotenoids production ($\mu\text{g g}^{-1}$), that is, total concentration of carotenoids (μg) in relation to the biomass of dried yeast. The carotenoids solutions used for antimicrobial and antioxidant action were crude extracts.

2.5. Determination of biomass and reducing sugars

After the extraction of carotenoids, the cells were washed with distilled water and centrifuged at $5000 \times g$ at $4^{\circ}\text{C}/10\text{ min}$. The biomass was quantified through drying at 65°C until it reached a constant mass. The procedure used for quantification of glucose was the DNS (3,5-dinitrosalicylic) methodology as described by Miller (1959) and it was measured by spectrophotometer (Biospectro, Model SP-220) at a wavelength of 540 nm.

2.6. β -carotene quantification

The β -carotene content was determined according to procedures described by Maldonado et al. (2008) with adaptations. The extracts were concentrated in nitrogen gas and were redissolved in 1 mL of methanol solution and filtered through a nylon membrane filter $0.22\text{ }\mu\text{m}$, (Millipore) before injection. Analyses were performed on High Performance Liquid Chromatograph (HPLC) (Shimadzu, Japan), equipped with quaternary pump LC-20AT model, diode array detector (DAD) EPDM-20A model, degasser DGU-20A5 model, interface CBM-20A model and automatic gun with autosampler SIL-20A model. The separations were carried out using A VP-ODS Shim-pack C18 column $25\text{ cm} \times 4.6\text{ mm} \times 5\text{ }\mu\text{m}$ and Shim-pack guard column GVP-ODS C18 column $10\text{ mm} \times 4.6\text{ mm} \times 5\text{ }\mu\text{m}$. The mobile phase was tetrahydrofuran: H_2O : methanol from 15:4:81 (v/v/v) to 30:0:70 (v/v/v) in a linear gradient in 55 min, maintaining this proportion until the end of the run. The flow rate was 0.6 mL min^{-1} . The column thermostat was set at 40°C , the injection volume was $20\text{ }\mu\text{L}$ and analyzed at 450 nm. The retention time of β -carotene extracted from *Rhodotorula mucilaginosa* CCMA 0156 was 32.487 min and quantified against peak area calibrations calculated from the standard curves of the β -carotene (1 mg mL^{-1}) standard (Sigma, USA).

2.7. Antioxidant activity

2.7.1. DPPH radical method

The antioxidant activity was determined by measuring the radical scavenging activity of DPPH (1,1-diphenyl-2-picrylhydrazyl) according to Brand-Williams and Berset (1995), with adaptations. The carotenoids solution (Item 2.4) (0.25 mL) was placed in test tubes to which was added 2.75 mL of the DPPH methanolic solution (0.06 mM). The tubes were left at rest in dark conditions. At intervals of 15, 30, 45 and 60 min, the absorbance of the tubes was measured at 515 nm. Synthetic antioxidant, butylated hydroxytoluene (BHT) (100 ppm), ascorbic acid (AA) (50 ppm) and β -carotene (100 ppm) were used as standard controls for the purpose of comparison. All analyses were performed in triplicate.

2.7.2. Co-oxidation system β -carotene/linoleic acid method

The antioxidant activity was evaluated by discoloration of the β -carotene/linoleic acid system as described by Marco (1968) and modified by Miller (1971). A 50 μ L aliquot of β -carotene solution [20 mg mL⁻¹ in chloroform (99%)] was placed in 250 mL Erlenmeyer flask filled in 40 μ L of linoleic acid, 1 mL chloroform, and 530 μ L tween 40. After homogenization, the chloroform was completely evaporated with nitrogen. After the initial reading (470 nm), the samples were kept at 40 °C in a water bath and the absorbance was measured after 2 h. The decrease in absorbance was compared to the control (without antioxidant). BHT, AA and Trolox were used as positive control.

2.8. Antibacterial activity

The antibacterial activity of the carotenoids was performed according to Gudiña et al. (2010), adapted. The evaluation of minimum inhibitory concentration (MIC) of carotenoids against pathogenic strains such as *Salmonella choleraesuis* ATCC 6539, *Staphylococcus aureus* ATCC 13565, *Pseudomonas aeruginosa* ATCC 15442, *Listeria monocytogenes* ATCC 19,117 and *Escherichia coli* ATCC 11,229 was determined by microdilution method in flat-bottom 96-well polystyrene microplates.

Suspensions of bacteria were prepared in saline solution (NaCl 0.85%) with the standard nephelometric McFarland scale of 1 ($\approx 8 \log$ CFU mL⁻¹). Different extracts of carotenoids concentrations were prepared: 0.00, 0.78, 1.56, 3.12, 6.25, 12.5, 25, 50% (v/v) in BHI broth in a final volume of 150 μ L per well. The bacterial inoculum ($8 \log$ CFU mL⁻¹) was added to the volume of 10 μ L per well. For all concentrations, wells were prepared without the bacterial inoculums, containing only extracts carotenoids diluted in the culture medium (BHI broth). Two wells did not contain extract of carotenoids and were used as activity and growth negative controls. Synthetic β -carotene (Sigma) at 1 mg mL⁻¹ was used as positive control. The microplates were covered and incubated at 37 °C/24 h. Triplicate assays were performed for all the carotenoids concentrations for each strain. The absorbance was measured at 600 nm in microplate reader (Multiskan GO - Thermo Scientific) prior to the incubation (time zero) and after the 24 h period. The growth inhibition percentages at different extracts of carotenoids concentrations for each microorganism were calculated:

$$\% \text{Growth inhibition}_c = [1 - (A_c/A_0)] \times 100$$

where A_c represents the absorbance of the well with a carotenoid concentration c and A_0 the absorbance of the control well (without carotenoid).

The minimum inhibitory concentration (MIC) was determined for each strain as the lowest concentration of carotenoid that completely inhibits measurable growth ($A_{600} = 0$). To establish the minimum bactericidal concentration (MBC), 10 μ L of each well with no visible growth were inoculated on plates containing TSA medium

(tryptone soya agar) (Himedia) incubated at 37 °C for 24 h. Plates that showed growth were considered to come from wells with bacteriostatic concentration of carotenoid.

2.9. Antifungal activity

Antifungal activity of the carotenoids was evaluated according to the methodology proposed by Lopes et al. (2013), with adaptations. Two strains of filamentous fungi aflatoxin producers, *Aspergillus flavus* (CCDCA 10520) and *A. parasiticus* (CCDCA 10510), and two ochratoxin producers strain, *A. ochraceus* (CCDCA 10612) and *A. carbonarius* (CCDCA 10608), were tested.

The fungi were grown for 7 days at 28 °C on plates containing Malt Extract Agar (MEA) (g L⁻¹: 20.0 of malt extract, 1.0 of bacteriological peptone, 10.0 glucose and 20.0 agar). After incubation, tween 80 (0.1%) was added to the solution over the entire plate surface and held with the rasp handle Drygalski to keep the spores suspension. The concentration of 10^5 spores mL⁻¹ (30 μ L) were added in 70 μ L carotenoid extract and incubated at 28 °C/2 h. After this period, 10 μ L was inoculated (microdrop) in MEA plates and incubated at 28 °C/7 days. For each fungus, negative controls were performed, in the absence of carotenoid. β -carotene synthetic (Sigma) (1 mg mL⁻¹) was used as positive control. The experiment was performed in triplicate.

3. Results and discussion

3.1. Characterization of raw materials

The coffee pulp and husks are naturally acidic, the pH in this study ranged from 4.5 to 4.7, in pulp extracts and coffee husk, respectively. After the pretreatment with KOH, the pH of the extracts were changed to 5.0 (pulp extract) and 5.7 (husk extract) (see Table 1). However, comparing the dry matter content in the samples in natura and pretreated there was a great loss of dry matter after pretreatment, as expected. The complex carbohydrates break after treatment thereby the total content should be increased. However, the pre-treatment was not efficient to solubilize the entire polysaccharide of the solid fraction.

There were no inhibitor compounds after pretreatment in the raw materials because the microbial growth did not affected (data not shown).

The coffee pulp and husk are the first by-products obtained during processing of the coffee fruit representing 29% and 12% of the dry weight of the berry, respectively (Murthy and Naidu, 2012). These by-products have similar chemical composition. Coffee pulp and husk are rich in carbohydrates (32–35%), proteins (7–15%) and minerals (10.7%) (Brand et al, 2001; Esquivel and Jiménez, 2012; Ulloa-Rojas et al, 2003;). Although the chemical composition of the coffee processing waste varies with the coffee variety, growing conditions, processing mode, in the developmental stage at which the grains were harvested (Pandey et al., 2000) the values for coffee pulp and coffee husk were found to be similar to that observed by other authors (Esquivel and Jiménez, 2012; Murthy and Naidu, 2012; Silva et al, 2013). The chemical composition of carbohydrates in the pulp and husks indicate that these residues are very promising for use in many biotechnological processes (Bonilla-Hermosa et al., 2014), solving the problem with the disposal of these residues. When the producing unit implements an Integrated Production Management System it may require ISO certification 14001. It means to prove to the market and society that the organization adopts a set of practices aimed at minimizing impacts that pose risks to the preservation of biodiversity which adds value to all products originating from that production unit (Nicoletta et al., 2004).

Table 1Physicochemical composition of pulp and coffee husk in nature and after pretreatment to carotenoids production by *R. mucilaginosa* CCMA0156.

Parameters	in nature		pretreated	
	Coffee pulp	Coffee husk	Coffee pulp	Coffee husk
pH	4.5	4.7	5.0	5.7
Dry matter (wt%)	81.64	98.23	24.21	30.50
Crude protein (wt%)	17.37	13.72	17.10	13.38
Carbohydrates (wt%)	15.68	13.61	12.01	10.07
Phenolic compounds (g 100 g ⁻¹)	2.62	1.0	nd	nd
Reducing sugar concentrations	9.63	13.5	9	13
Humidity (%)	100.0	91.0	22.5	28.5
Lipid content (%)	1.2	0.5	nd	nd

Nd = Non determined.

3.2. Production of carotenoids

From the fermentation media containing coffee husk and pulp coffee extracts could obtain total and specific production of carotenoids as well as biomass *Rhodotorula mucilaginosa* CCMA 0156 (see Table 2). Most production of carotenoid and biomass were in the husk extract media fermentation.

Yeast had carotenoids production behavior which was similar in both fermentation media. The carotenoids production was accompanied by a consumption of reducing sugars, biomass, increase in pH values and evolution of carotenoid production (see Fig. 1). Although the initial reducing sugar content in husk extract (HE) is greater than in the pulp extract (PE), because the composition of media, *Rhodotorula mucilaginosa* CCMA 0156, the consumption of sugars was near 86.8% and 84%, respectively.

The pH of the media (coffee extract pulp and husk extract) had increased during the production of carotenoids. These changes in pH are due to the growth of yeast. It is known that the pH decreases during the first hours of fermentation and increases during the accumulation of carotenoids and then remains constant until the end time (Frengova and Beshkova, 2009).

According to our study (Fig. 1), during the fermentation time, population growth was not expressive, i.e., the production of pigment was not related to growth. This indicates that the carotenoids are secondary products of metabolism of yeasts, in which the accumulation of these pigments starts at the end of the exponential growth phase and continues during the stationary phase, perhaps because at this stage a large number of cells are unviable and growth rate approximates to zero (Valduga et al., 2008, 2009). This is due to the regulation of enzymes involved in response to stress during production of the carotenoids stationary phase; there is an increase in the levels of expression of catalase (Barbachano-Torres et al., 2014). According to Martinez-Moya et al. (2015), the greatest abundance of enzymes involved in response to stress in stationary phase would be related to the induction of carotenoid production. Thus, the maximum biomass and accumulation of carotenoids is rarely achieved at the same time.

As well as our results, Petrik et al. (2014) showed that coffee processing wastes are potential substrates for economical production of pigments by yeast. However, compared to the different

wastes, the use of coffee pulp and husk as a carbon source resulted in increased production of carotenoids by the yeast *R. mucilaginosa* CCMA 0156 (16.36 ± 0.073 mg L⁻¹ coffee pulp and husk extract was 21.35 ± 0.067 mg L⁻¹) rather than spent coffee grounds used by *Sporobolomyces roseus* (12.59 mg L⁻¹) (Petrik et al., 2014). Our results were also 6–8x higher than those obtained by Cheng and Yang (2016), that reported a 2.61 mg L⁻¹ production by the yeast *R. mucilaginosa* in cane molasses. This advantage is due to the fact that the cane molasses sugar presents impurities contained in the midst causing critical problems during fermentation, such as inhibition of microbial growth, influence on the pH of the substrate and the means of fermentation and are involved in the inactivation of enzymes associated with the biosynthesis of the product. In this way, there is a possibility that the same species of yeasts can present different behaviors when grown in different substrates, due to induction of substrate, i.e., the metabolism can be directed to a greater or lesser production of carotenoids (Aksu and Eren, 2005; Cheng & Yang, 2016; Maldonado et al., 2008). Therefore, the best selection, especially carbon sources derived from residues, is important for the development of certain criteria, which can be used for an assessment in terms of their suitability to facilitate the production of carotenoids.

3.3. β -carotene quantification

β -carotene was produced by *R. mucilaginosa* CCMA 0156 in the two culture media used, but, in different concentrations. The β -carotene quantification from total carotenoids produced in coffee pulp extract and husk extract media are shown in Table 2. The major β -carotene production received from coffee husk extract (97,560 ± 51.8 μ g L⁻¹) being 4.57% into total carotenoids. The difference in the β -carotene production from different substrates may be related to composition of the coffee pulp extract and husk extract. The β -carotene production percentage was higher in husk extract. The pulp and husk added in the media had different sugar concentration. Other parameters as nitrogen sources, was not altered during all cultivation. High β -carotene content was produced by mutant of *Rhodotorula glutinis* grown in media containing ammonium nitrate, soy peptone or peptone the nitrogen sources (Bhosale and Gadre 2001). On the other hand, cultivation of *Rhodo-*

Table 2Concentration of carotenoids (specific, total carotenoids and β -carotene) and biomass obtained by *Rhodotorula mucilaginosa* CCMA 0156 in coffee pulp extract (PE) and coffee husk extract (HE) and in artificial media culture (AMC).

Waste	Specific carotenoids (μ g g ⁻¹)	Total carotenoids (μ g L ⁻¹)	β -carotene (μ g L ⁻¹)	β -carotene (%)	Biomass (g L ⁻¹)	Ref.
ACM	69.0	590.69 ± 0.5	139.16 ± 0.2	nd	8.6 ± 0.20	Maldonado et al., 2008
PE	2065.94 ± 42.0	16362.27 ± 73.0	670.52 ± 20.6	4.10	7.92 ± 0.8	This work
HE	2457.81 ± 31.6	21358.40 ± 67.2	975.60 ± 51.8	4.57	8.69 ± 0.3	This work
Whey milk	140.50 ± 29.14	nd	nd	nd	6.78 ± 1.88	Marova et al., 2012
Cane molasses	268.6	2611.0	nd	23.8	9.72	Cheng & Yang, 2016

Nd = non determined.

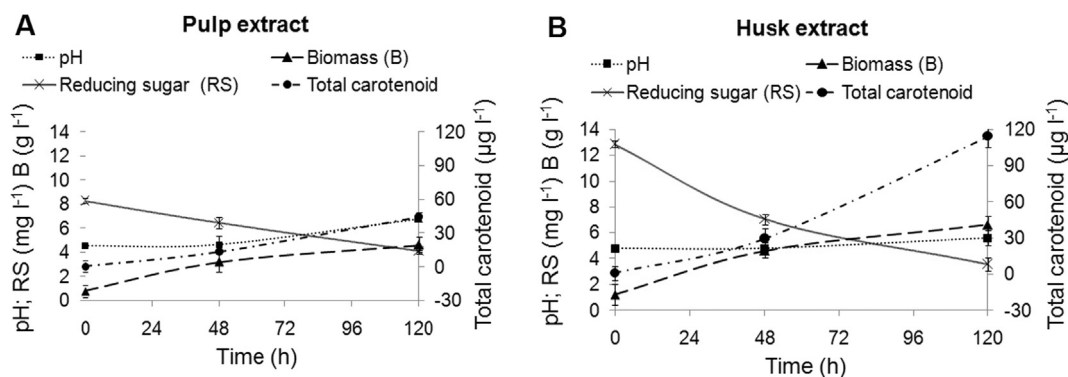


Fig. 1. Reducing sugars, pH evolution, biomass and total carotenoid in coffee pulp extract (A) and husk extract (B) during the production of carotenoids by *Rhodotorula mucilaginosa* CCMA 0156, at 28 °C, 160 rpm in the dark for 120 h. The bars indicate standard deviation.

torula glutinis var. *glutinis* in medium containing ammonium nitrate resulted in the highest percentage of β -carotene, while peptone gave the highest percentage of torulene (El-Banna et al., 2012). However, in this work, the difference over relative production of β -carotene, between the media tested, is insignificant, so in this case the better choice of composition media should be the lowest cost, i.e., the coffee husk extract media.

Other parameters influence carotenoids production. Aksu and Eren, 2005; Maldonado et al. (2008); Cheng and Yang (2016) reported that different incubation temperatures may alter the number of carotenoids. Temperatures between 20° and 30 °C are favorable to the growth of *Rhodotorula*. In our study, the production of carotenoids was performed at 28 °C and according to what is described by Cheng and Yang (2016), temperatures close to 31 °C, do not contribute to the synthesis of β -carotene and torulene, but favor the torularodine. This fact may explain the low amount of β -carotene (4.5%) found in our study. Considering that, the coffee production is carried out in tropical regions of the world, temperatures between 20 and 32 °C is very common. The airflow rate is also an important factor for substrate assimilation and consequent synthesis of carotenoids, since intensive aeration stimulates the synthesis of β -carotene (Simova et al., 2003), but the air supply is more expensive than agitation. However, thinking about the use of waste is supposed that the production method should be simpler and cheaper if possible and, was observed that, the use of coffee pulp and coffee husk was competitive and economically viable. The commercial value, per gram, of synthetic β -carotene is around \$ 10.40 (Sigma-Aldrich, 2018)). In contrast, the cost to produce 1 g of pigment in this study was \$ 5.04.

The β -carotene production by *R. mucilaginosa* CCMA 0156 was higher than carotenoid production by 4 strains *R. mucilaginosa* as reported by Maldonado et al. (2008) from Brazilian soil. Yeast strains in artificial media culture showed that strains genus *Rhodotorula mucilaginosa*-12, *Rhodotorula mucilaginosa*-137, *Rhodotorula mucilaginosa*-108 and *Rhodotorula mucilaginosa*-135 produced 105 $\mu\text{g L}^{-1}$, 139 $\mu\text{g L}^{-1}$, 129 $\mu\text{g L}^{-1}$ and 95 $\mu\text{g L}^{-1}$ of β -carotene, respectively.

3.4. Antioxidant activity

Analysis of antioxidant activity using the radical DPPH assay and the co-oxidation of β -carotene against linoleic acid assays have long been used to verify the scavenging capacity of free radicals in many natural products (Li et al., 2011). Comparing both detection methods in relation to the antioxidant activity, the co-oxidation of the β -carotene against linoleic acid was more effective in the analysis of carotenoids produced by the yeast extracts than the DPPH method. Once carotenoids are composed of lipophilic nature, they

have a greater affinity for this method (Kuskoski et al., 2005). Carotenoids produced in the media with coffee pulp extract showed the best antioxidant protection value ($98.26 \pm 1.2\%$) compared to the carotenoids produced in the coffee husk extract ($64.13 \pm 5.2\%$) (see Table 3). The antioxidant efficiency varies between different carotenoids from media with different conditions (Squina and Mercadante, 2003) using the same yeast strain (Damodaran et al., 2008). Thus, we had different types of carotenoids produced by *R. mucilaginosa* CCMA 0156 in coffee pulp extract from those produced in coffee husk extract. According to Mata-Gómez et al. (2014), Tinoi et al. (2005) and Squina and Mercadante (2003), the main carotenoids produced by *Rhodotorula* are β -carotene, torulene e torularodine. The protection potential of the carotenoids produced in coffee pulp extract were similar to those found for BHT (85.9%), lipophilic synthetic antioxidant and higher than Trolox and AA (hydrophilic), 73.7 and 70.6%, respectively (see Table 3). Manimala and Murugesan (2014) reported carotenoid antioxidant activity produced by yeast *Sporobolomyces* sp (75.04%), using DPPH method, similar to our results. The antioxidant action in yeast carotenoids proves that in the microbial cell, the pigment age as protection over induced oxidative damage (Naghavi et al., 2015). The same action is observed in human cells preventing the formation of reactive products and neutralization of organism damage caused by free radicals. Thus, they are used in the prevention and treatment of chronic diseases such as cancer, diabetes, cardiovascular and autoimmune disorders (Vachali et al., 2012).

3.5. Antimicrobial activity

The antimicrobial activity of the total carotenoids isolated from *Rhodotorula mucilaginosa* CCMA 0156 was determined by measuring the growth inhibition percentages obtained for some patho-

Table 3

Antioxidant activity of carotenoids extracts produced by *Rhodotorula mucilaginosa* CCMA 0156 from the different culture media (coffee pulp extract – PE and husk extract – HE) under different methods (β -carotene/linoleic acid) and (DPPH).

Sample	% Protection β -carotene/linoleic acid	% Sequestration DPPH
PE	98.26 ± 1.2	72.94 ± 2.7
HE	64.13 ± 5.2	76.74 ± 1.2
TROLOX	73.70 ± 1.4	–
BHT	85.92 ± 2.9	90.82 ± 1.2
AA	70.63 ± 1.48	96.14 ± 0.8
β -carotene	–	52.24 ± 2.5

PE – pulp extract; HE – husk extract. Synthetic antioxidants: Trolox – (6-hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid) BHT – Butylated hydroxytoluene; AA – ascorbic acid and β -carotene. DPPH – 1,1 – diphenyl – 2 – picrilidrazil.

genic bacteria (Fig. 2). From those results, the MIC for each microorganism was determined. Furthermore, whenever possible, the MBC was also determined. All total carotenoids were effective against all the microorganisms tested, although to different degrees.

For all the microorganisms tested, antimicrobial activity was observed even at low carotenoid concentrations, and a complete growth inhibition was achieved for the 4 microorganisms at the highest carotenoid concentration assayed (50 v/v), 1.225 $\mu\text{g } \mu\text{L}^{-1}$ (PE) and 1.600 $\mu\text{g } \mu\text{L}^{-1}$ (HE). *Salmonella*, *E. coli*, *Staphylococcus aureus* and *Listeria* exhibit complete inhibition of growth by the action of carotenoid extract through concentrations (MIC) between 25 and 50 (v/v) (0.612–1.225 $\mu\text{g } \mu\text{L}^{-1}$).

Furthermore, it is noticeable that even when MIC and MBC did not occur at the same time, a high growth inhibition was observed from 73.2 to 89.9% (HE), with the highest carotenoid concentration assayed (25 v/v), 0.800 $\mu\text{g } \mu\text{L}^{-1}$. Synthetic β -carotene (1000 $\mu\text{g } \mu\text{L}^{-1}$) used as positive control, was able to inhibit the growth of most bacteria tested but had no action on the growth of *Listeria monocytogenes* ATCC 19117. This inhibition is important because these bacteria are responsible for food contamination and may cause dysentery, vomiting and even death.

The majority of the studies in the literature on antimicrobial potential of pigments are related to bacterial pigments or filamentous fungi. However, in a study similar to ours, Manimala and Murugesan (2014) evaluated the antimicrobial activity of the carotenoid produced by the yeast *Sporobolomyces* sp. Among the eval-

uated pathogens, the pigment was found to be more effective against *E. coli* and *S. aureus*. Lopes et al. (2013) evaluated the antimicrobial activity, the pigment produced by *Penicillium chrysogenum* IFL1. Predominantly, inhibition occurred against bacteria *Staphylococcus aureus* (25%), *Pseudomonas aeruginosa* (50%) e *Bacillus cereus* (25%) and phytopathogenic fungi *Fusarium oxysporum* (50%). Endophytic fungal pigment was found to be superior to the commercial antibiotic Streptomycin against human pathogenic bacteria (Visalakshi & Muthumary, 2010). In our study, a low MIC value between coffee pulp extract (0.612–1.225 $\mu\text{g } \mu\text{L}^{-1}$) and husk extract (0.800–1.600 $\mu\text{g } \mu\text{L}^{-1}$) was able to inhibit the growth of pathogenic bacteria tested. These values were lower than those reported by Kim et al. (2006) who evaluated the antimicrobial potential of *Monascus* pigments against several strains of bacteria and fungi. The control red pigment exhibited minimal inhibitory concentration (MIC) values, higher than 32 mg mL^{-1} . Our study suggests as a new alternative, the use of carotenoids produced by yeast with potential antimicrobial activity.

Interesting preliminary results were observed regarding the antifungal potential of crude extracts of carotenoids. All extracts tested were able to completely inhibit the growth of all strains of *Aspergillus* evaluated (data not shown). The results suggest that these carotenoids could be used to control the producer mycotoxin fungi in food, for example to add the extracts of carotenoids in coffee beans as biological control. However, future studies should be performed to determine the minimum inhibitory concentration (MIC) of these extracts against these fungi as well as testing effi-

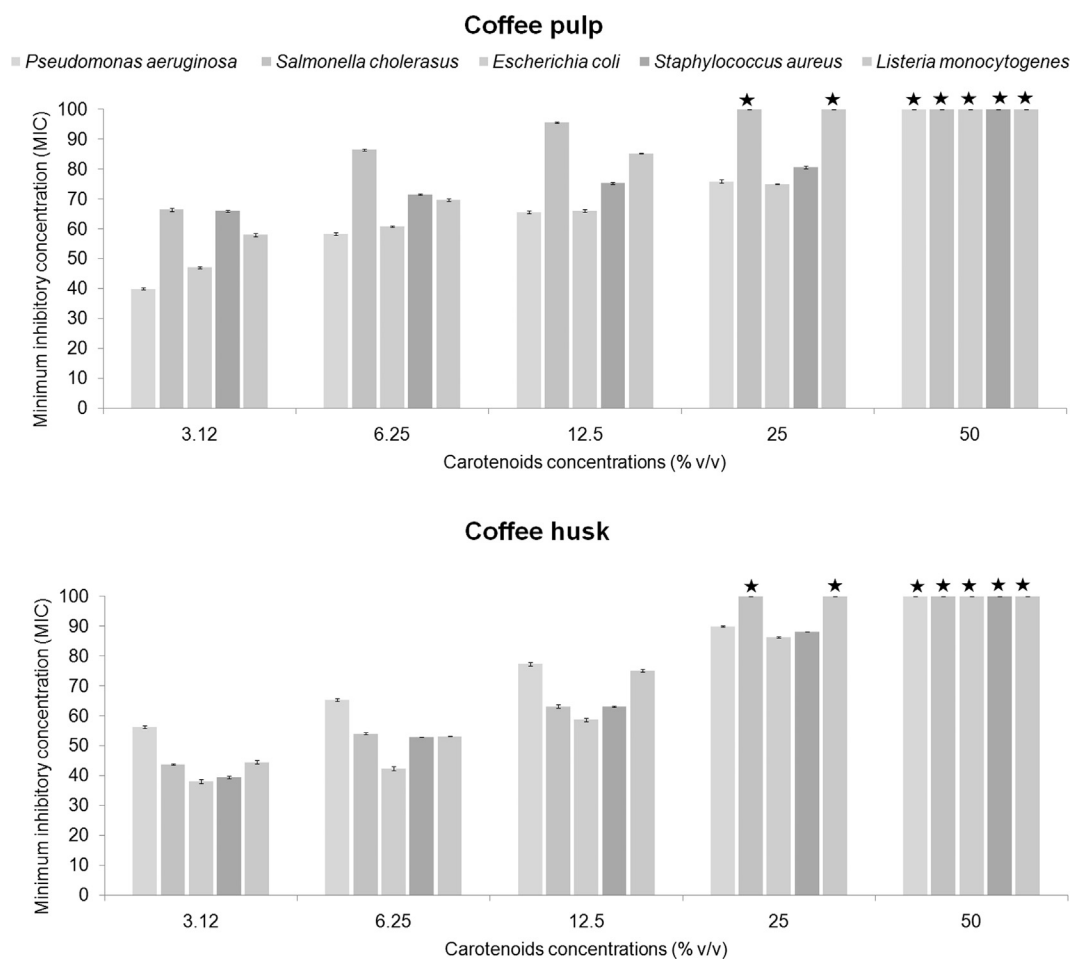


Fig. 2. Percentage inhibition of growth (MIC) of pathogenic bacteria by the action of carotenoids extracts recovered from *Rhodotorula mucilaginosa* CCMA 0156 in different fermentation medium in the different tested concentrations (coffee pulp: 3.12/0.076 $\mu\text{g } \mu\text{L}^{-1}$; 6.25/0.153 $\mu\text{g } \mu\text{L}^{-1}$; 12.5/0.306 $\mu\text{g } \mu\text{L}^{-1}$; 25/0.612 $\mu\text{g } \mu\text{L}^{-1}$; 50/1.225 $\mu\text{g } \mu\text{L}^{-1}$) and coffee husk: 3.12/0.100 $\mu\text{g } \mu\text{L}^{-1}$; 6.25/0.200 $\mu\text{g } \mu\text{L}^{-1}$; 12.5/0.400 $\mu\text{g } \mu\text{L}^{-1}$; 25/0.800 $\mu\text{g } \mu\text{L}^{-1}$; 50/1.600 $\mu\text{g } \mu\text{L}^{-1}$. ★ Indicates the minimum bactericidal concentration (MBC).

ciency of these carotenoids against other species of pathogenic fungi. Future studies on mechanisms of action of carotenoids extracts produced by *Rhodotorula mucilaginosa* CCMA 0156 are necessary to understand the antimicrobial mode of action of these pigments.

4. Conclusions

From the cheap raw material used in the fermentation medium, it can be noted that the strain *R. mucilaginosa* CCMA 0156 is a strong candidate for the production of carotenoids. The culture medium that provided the highest production of total carotenoids contained coffee husks while the highest antioxidant and antimicrobial power was observed in carotenoids from pulp coffee extract. These biological activities obtained from carotenoids allow use it as a coloring substance that presents antioxidant properties, adding value to the product. These biological activities are probably the other carotenoids different to β carotene because the concentration in this condition was low. However, the cytotoxicity tests should be performed prior to commercial use. In addition, few studies have demonstrated the biological properties of the carotenoids could be used in food as natural additive, cosmetic industries or biological control.

Acknowledgments

We thank the Culture Collection of Microorganisms of Agricultural Microbiology (CCMA) and the Culture Collection of the Department of Food Sciences (CCDCA) of the Federal University of Lavras - UFLA for supplying the strains used in this study and CNPq and CAPES to financial support.

References

- Aksu, Z., Eren, A.T., 2005. Carotenoids production by the yeast *Rhodotorula mucilaginosa*: use of agricultural wastes as carbon source. *Process. Biochem.* 40, 2985–2991.
- Association of official analytical chemists – AOAC, 1995. Official methods of analysis of the association of official analytical chemists: 16. ed. Washington.
- Barbachano-Torres, A., Castelblanco-Matiz, L.M., Ramos-Valdivia, A.C., Garcia-Rojas, C.M., 2014. Analysis of proteomic changes in colored mutants of *Xanthophyllomyces dendrorhous* (Phaffia rhodozyma). *Arch. Microbiol.* 196, 411–421.
- Bhosale, P.B., Gadre, R.V., 2001. Production of β -carotene by a mutant of *Rhodotorula glutinis*. *Appl. Microbiol. Biotechnol.* 55, 423–427.
- Bonilla-Hermosa, V.A., Duarte, W.F., Schwan, R.F., 2014. Utilization of coffee by-products obtained from semi-washed process for production of value-added compounds. *Bioresour. Technol.* 166, 142–150.
- Brand, D., Pandey, A., Rodríguez-León, J.A., Roussos, S., Brand, I., Soccol, C.R., 2001. Packed bed column fermenter and kinetic modeling for upgrading the nutritional quality of coffee husk in solid-state fermentation. *Biotechnol. Prog.* 17, 1065–1070.
- Brand-Williams, W., Cuvelier, M.E., Berset, C., 1995. Use of free radical method to evaluate antioxidant activity. *Lebensm. Wiss. Technol.* 28, 25–30.
- Cheng, Y.T., Yang, C.F., 2016. Using strain *Rhodotorula mucilaginosa* to produce carotenoids using food wastes. *J. Taiwan Inst. Chem. Eng.* 61, 270–275.
- Damodaran, S., Parkin, K., Fennema, O.R., 2008. Fennema's Food Chemistry. CRC Press, Boca Raton, p. 1144.
- Davies, B.H., 1976. Carotenoid. In: Goodwin, T.W. (Ed.), *Chemistry and Biochemistry of Plant Pigments*. Academic Press, New York, pp. 38–165.
- Dubois, M., Gilles, K.A., Hamilton, J.K., Rebers, P.A., Smith, F., 1956. Calorimetric method for determination of sugars and related substances. *Anal. Chem.* 28, 350–356.
- El-Banna, A.A., El-Razek, A.M.A., El-Mahdy, A.R., 2012. Some factors affecting the production of carotenoids by *Rhodotorula glutinis* var. *glutinis*. *Food. Nutr. Sci.* 3, 64–71.
- Esquivel, P., Jiménez, V.M., 2012. Functional properties of coffee and coffee byproducts. *Food. Res. Int.* 46, 488–495.
- Frengova, G.I., Beshkova, D.M., 2009. Carotenoids from *Rhodotorula* and *Phaffia*: yeasts of biotechnological importance. *J. Ind. Microbiol. Biotechnol.* 36, 163–180.
- Gudiña, E.J., Rocha, V., Teixeira, J.A., Rodrigues, L.R., 2010. Antimicrobial and antiadhesive properties of a biosurfactant isolated from *Lactobacillus paracasei* ssp. *paracasei* A20. *Lett. Appl. Microbiol.* 50, 419–424.
- Kaiser, P., Surmann, P., Vallentin, G., Fuhrmann, H., 2007. A small-scale method for quantitation of carotenoids in bacteria and yeasts. *J. Microbiol. Methods.* 70, 142–149.
- Kim, C., Jung, H., Kim, Y.O., Shin, C.S., 2006. Antimicrobial activities of amino acid derivatives of *Monascus* pigments. *FEMS. Microbiol. Lett.* 264, 117–124.
- Kirti, K., Amita, S., Priti, S., Kumar, A.M., Jyoti, S., 2014. Colorful world of microbes: carotenoids and their applications. *Adv. Biol.* 2014, 13.
- Kuskoski, E.M., Asuero, A.G., Troncoso, A.M., Mancini, F.J., Fett, R., 2005. Aplicación de diversos métodos químicos para determinar actividad antioxidante en pulpa de frutos. *Cienc. Tecnol. Aliment.* 25, 726–732.
- Li, P., Huo, L., Su, W., Lu, R., Deng, C., Liu, L., Deng, Y., Guo, N., Lu, C., He, C., 2011. Free radical-scavenging capacity, antioxidant activity and phenolic content of *Pouzolzia zeylanica*. *J. Serb. Chem. Soc.* 76, 709–717.
- Lopes, F.C., Tichota, D.M., Sauter, I.P., Meira, S.S.M., Segalin, J., Rott, M.B., Rios, A.O., Brandelli, A., 2013. Active metabolites produced by *Penicillium chrysogenum* FL1 growing on agro-industrial residues. *Ann. Microbiol.* 63, 771–778.
- Machado, C.M.M., Soccol, C.R., Oliveira, B.H., Pandey, A., 2002. Gibberellic acid production by solid-state fermentation in coffee husk. *Appl. Biochem. Biotechnol.* 15, 102–106.
- Maldonado, I.R., Rodríguez-Amaya, D.B., Scamparini, A.R.P., 2008. Carotenoids of yeasts isolated from the Brazilian ecosystem. *Food. Chem.* 107, 145–150.
- Malik, K., Tokkas, J., Goyal, S., 2012. Microbial Pigments: a review. *Int. J. Microb. Res. Technol.* 1, 361–365.
- Manimala, M.R.A., Murugesan, R., 2014. In vitro antioxidant and antimicrobial activity of carotenoid pigment extracted from *Sporobolomyces* sp. isolated from natural source. *J. Appl. Nat. Sci.* 6, 649–653.
- Marco, G.J., 1968. A rapid method for evaluation of antioxidants. *J. Am. Oil. Chem. Soc.* 45, 594–598.
- Marova, I., Carnecka, M., Halienova, A., Certik, M., Dvorakova, T., Haronikova, A., 2012. Use of several waste substrates for carotenoid-rich yeast biomass production. *J. Environ. Manage.* 95, 338–342.
- Martinez-Moya, P., Niehaus, K., Alcaíno, J., Baeza, M., Cifuentes, V., 2015. Proteomic and metabolomic analysis of the carotenogenic yeast *Xanthophyllomyces dendrorhous* using different carbon sources. *BMC Genom.* 16, 289.
- Mata-Gómez, L.C., Montañez, J.C., Méndez-Zavala, A., Aguilar, C.N., 2014. Biotechnological production of carotenoids by yeasts: an overview. *Microb. Cell. Fact.* 13, 1–12.
- Miller, G.L., 1959. Use of Dinitrosalicylic Acid Reagent for Determination of Reducing Sugar. *Anal. Chem.* 31, 426–428.
- Miller, H.E., 1971. A simplified method for the evaluation of antioxidant. *Journal of the American Oil Chemists' Society.* 48, 91.
- Murthy, P.S., Naidu, M.M., 2012. Sustainable management of coffee industry by-products and value addition—a review. *Resour. Conserv. Recy.* 66, 45–58.
- Naghavi, F., Hanachi, P., Sodi, M.R., Sabora, A., 2015. The capability of *Rhodotorula looffiae* to produce carotenoid. *ZJRMS.* 17, 52–56.
- Nasri, H., Baradaran, A., Shirzad, H., Rafeian-Kopaei, M., 2014. New concepts in nutraceuticals as alternative for pharmaceuticals. *Int. J. Prev. Med.* 5, 1487–1499.
- Nicolella, G., Marques, J.F., Skorupa, L.A., 2004. Sistema de Gestão Ambiental: aspectos teóricos e análise de um conjunto de empresas da região de Campinas. SP, Jaguariúna, Embrapa Meio Ambiente, p. 42.
- Pandey, A., Soccol, C.R., Nigam, P., Brand, D., Mohan, R., Roussos, S., 2000. Biotechnological potential of coffee pulp and coffee husk for bioprocesses. *Biochem. Eng. J.* 6, 153–162.
- Petrik, S., Obrucá, S., Benesová, P., Márová, I., 2014. Bioconversion of spent coffee grounds into carotenoids and other valuable metabolites by selected red yeast strains. *Biochem. Eng. J.* 90, 307–315.
- Sachindra, N.M., Bhaskar, N., Mahendrak, N.S., 2006. Recovery of carotenoids from shrimp waste in organic solvents. *Waste Manage.* 26, 1092–1098.
- Sigma-Aldrich, 2018. <<https://www.sigmaaldrich.com>>, acessada em Maio.
- Silva, C.F., Vilela, D.M., Cordeiro, C.S., Duarte, W.F., Dias, D.R., Schwan, R.F., 2013. Evaluation of a potential starter culture for enhance quality of coffee fermentation. *World. J. Microbiol. Biotechnol.* 29, 235–247.
- Simova, E.D., Frengova, G.I., Beshkova, D.M., 2003. Effect of aeration on the production of carotenoid pigments by *Rhodotorula rubra* and *Lactobacillus casei* subsp. *casei* co-cultures in whey ultrafiltrate. *Z. Naturforsch.* 58c, 225–229.
- Squina, F.M., Mercadante, A.Z., 2003. Análise, por CLAE, de carotenóides de cinco linhagens de *Rhodotorula*. *Braz. J. Pharm. Sci.* 39, 309–318.
- Tinoi, J., Rakariyatham, N., Deming, R.L., 2005. Simplex optimization of carotenoid production by *Rhodotorula glutinis* using hydrolyzed mung bean waste flour as substrate. *Process. Biochem.* 40, 2551–2557.
- Ulloa-Rojas, J.B., Verreth, J.A.J., Amato, S., Huismans, E.A., 2003. Biological treatments affect the chemical composition of coffee pulp. *Bioresour. Technol.* 89, 267–274.
- P. Vachali P., Bhosale P.S., Bernstein. 2012. Microbial carotenoids. In: José-Luis Barredo (Ed.), *Microbial Carotenoids from fungi: methods and protocols Methods. Mol. Biol.* vol. 898, pp. 41–59.
- Valduga, E., Valerio, A., Tatsch, P.O., Treichel, H., Furigo, A.J., Di Luccio, M., 2009. Assessment of cell disruption and carotenoids extraction from *Sporidiobolus salmonicolor* (CBS 2636). *Food. Bioproc. Technol.* 2, 234–238.
- Valduga, E., Valerio, A., Treichel, H., Di Luccio, M., Furigo, A.J., 2008. Study of the bio-production of carotenoids by *Sporidiobolus salmonicolor* (CBS 2636) using pre-treated agro-industrial substrates. *J. Chem. Technol. Biotechnol.* 83, 1267–1274.
- Visalakshi, S., Muthumary, J., 2010. Taxol (anticancer drug) producing endophytic fungi: an overview. *Int. J. Pharma. Bio. Sci.* 1, 1–9.